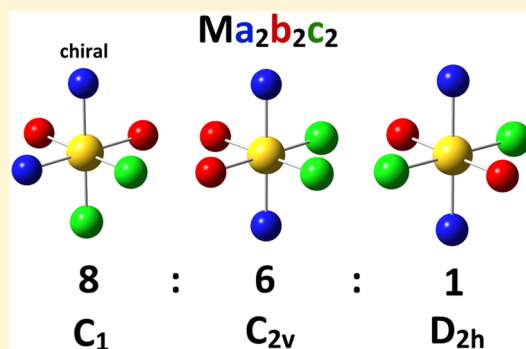


Stereoisomerism in Lanthanide Complexes: Enumeration, Chirality, Identification, Random Coordination Ratios

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Supporting Information

ABSTRACT: The concept of random coordination ratios, RCRs, is advanced for lanthanide complexes. RCRs describe the relative probabilities of occurrence of subsets of stereoisomers of same-symmetry point groups in the limiting situation when energetic effects are equivalent. We then introduce a method to uniquely identify the stereoisomer of the coordination polyhedron of a given crystallographic structure and introduce a notation that fully characterizes its stereochemistry in an unambiguous manner, from which absolute configuration naturally follows. De facto, the coordination chirality in lanthanide complexes is a frequently overlooked property, even though these compounds often exhibit, when luminescent, high dissymmetry factors. With our methodology, we even managed to recognize a known dilanthanide complex as a meso compound, with both metal ions functioning as stereogenic centers. To achieve these results, we enumerate all possible stereoisomers of lanthanide complexes with coordination numbers from 4 to 9 for all combinations of monodentate, symmetric and asymmetric bidentate ligands, and for several shapes of coordination polyhedra. We confirmed the number of stereoisomers for each case by means of Pólya's theorem. We further classified all stereoisomers according to their symmetry point groups and generated their Cartesian coordinates. This collection of all coordination polyhedra stereoisomer geometries, which is made available in the [Supporting Information](#), can also be used to easily build starting-point geometries for theoretical calculations of metal complexes.



INTRODUCTION

Lanthanide complexes are known to display a variety of coordination numbers, as well as ligand arrangements around the central trivalent metal ion with little preference for specific bond directions.^{1,2} This is because the trivalent lanthanide ions appear to the ligands as essentially charged spherically symmetric electron densities. As a result, the chemical bonds between the lanthanide ion and its ligands are fundamentally electrostatic, with the ligands being attracted by the central metal ion, while accommodating the steric interactions among themselves.

This bonding simplicity led to the development of the quantum-chemical Sparkle Model.³ This simplicity also facilitates, for example, the employment of density functional approaches⁴ without the explicit inclusion of the inner 4f subshell, all aiming at addressing the various issues related to the numerous properties, leading, for example, to bioanalytical and biomedical applications.⁵ Other applications of lanthanide complexes include spectroscopy,¹ self-assembly,⁶ nanoparticles,⁷ polymers,⁸ luminescence,⁹ circularly polarized luminescence,¹⁰ magnetic hysteresis,¹¹ magnetic resonance imaging (MRI),¹² medicine,¹³ etc.

However, beneath this bonding simplicity, there lies a hidden complexity, largely unacknowledged in both theoretical

and experimental studies of lanthanide complexes: stereoisomerism and, within it, coordination chirality.

Consider, for example, the well-known eight coordination complex tetrakis(6,6,6-trifluoro-2,2-dimethyl-3,5-hexanedionato)-cerium(III), $[Ce(fdh)_4]^-$.¹⁴ If a scientist wants to compute its structure with any quantum-chemical method, a starting geometry for this complex must be assumed. As will be shown in this article, in the absence of any experimental evidence, there will be at least 349 very different and legitimate starting geometries that can be considered for $[Ce(fdh)_4]^-$. Which one of these 349 possible starting geometries should the scientist use for his calculations? Indeed, without a systematized procedure, it would be utterly impossible to generate by hand and with complete certainty all 349 starting geometries for the theoretical calculations, let alone choose, among them, the relevant ones. For example, which one of them would be closer to the experimentally obtained geometry in a crystal for a given synthetic and crystallization procedure? And which ones of them might coexist in a given solution, in a situation that could be comparable to an alphabet soup? The fact is that the experimentally obtained crystal structures of $NH_4[Ce(fdh)_4]$ revealed two “similar but not identical”¹⁴ complex

Received: April 24, 2018



stereoisomers in the same unit cell. In addition, crystals of $\text{Na}[\text{Ce}(\text{fdh})_4]$ revealed a third different stereoisomer.¹⁴ The existence of these three different and well-characterized stereoisomers of $[\text{Ce}(\text{fdh})_4]$ is one of the many indications of the fact that stereoisomerism complexity, involving both diastereoisomerism and optical isomerism, is truly a phenomenon that cannot be ignored in lanthanide coordination compounds.

In this Article, Sections I to V introduce the concepts behind the new indices we developed: random coordination ratios and random coordination weights; Section VI presents enumeration, indices, point-group symmetries, and chirality in tabular form, as well as geometries, for all coordination polyhedra with coordination numbers from 4 to 9 for the most important coordination polyhedra shapes; Sections VII and VIII introduce a new notation coupled to an efficient technique for the complete characterization of the stereochemistry of an arbitrary coordination polyhedron, useful for crystallography; Sections IX to XII address experimental evidence of the random coordination ratios and coordination chirality, including the identification of an existing crystallographic structure of a dilanthanide as a meso compound; finally, Section XIII discusses the applicability of these results to theoretical computational chemistry.

RESULTS AND DISCUSSION

I. Random Coordination Ratios. When there are repeating equivalent atoms directly coordinated to the metal ion in a complex, random coordination ratios emerge. These ratios disclose why one complex configuration is more likely to occur than another, without taking into account energetic effects. To better understand this, let us first consider the simple case of a square planar complex, such as $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$. If one considers all possibilities of adding two amino and two chloride groups around the platinum ion, six different configurations will be obtained, four cis and two trans (Figure 1). All four cis C_{2v} configurations represent the same compound and are equivalent in the sense that all four can be transformed into one another by rotations around the central metal ion. The same happens for the two trans D_{2h} configurations, implying that the cis configuration would be 2 times more frequent than the trans configuration if

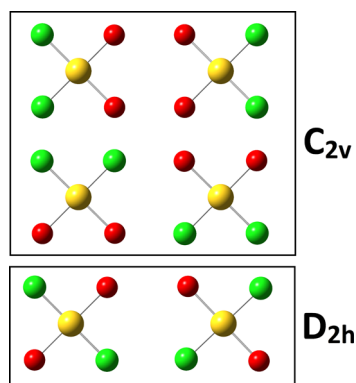


Figure 1. All six possible configurations that are obtained by adding two pairs of identical ligands (red and green) to a central metal (yellow) in the square planar shape, SP-4. Four of them will be cis, of point group C_{2v} , and two will be trans, of point group D_{2h} . Note that both D_{2h} structures can be transformed into one another by rotations. The same is true for the four C_{2v} structures.

coordination happened randomly and no differentiating energetic effects were present. Note, in Figure 1, that as symmetry is increased from C_{2v} for the cis compound, to D_{2h} for the trans compound, the number of equivalent configurations decreases. In performing this work, we detected that the configuration with the largest number of rotations of its symmetry point group will always be the one with the smallest number of equivalent configurations, and thus will be the least likely to be obtained by random coordination when energetic effects are not considered.

This proportion of 2:1, for a square planar complex with two pairs of identical ligands, is what we call a random coordination ratio, RCR. Of course, random coordination ratios team up with energetic effects, and therefore their knowledge is necessary, for example, for the correct interpretation of both frequencies of occurrence of structures, as well as synthetic reaction yields.

Let us now use this same complex to exemplify the methodology we developed in this article. As such, complex $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ is recognized as belonging to the group of complexes of coordination number 4, square planar (SP-4), and of general formula Ma_2b_2 , where M stands for a metal and “a” and “b” are different monodentate ligands. First, we assume all ligands are different and generate all Mabcd structures that cannot be transformed into each other by rotations. The number of such structures can be easily calculated as being the number of permutations of the ligands, $4!$, divided by the number of proper rotations of the square planar ideal structure, which in this case is eight, that is identity plus two C_4 and one C_2 axes perpendicular to the molecular plane, and four C_2 axes in the molecular plane. Therefore, the resulting number of structures that cannot be transformed into one another by rotations is $4!/8 = 3$. These three structures are thus all equally probable and form the set of stereoisomers for the generic formula Mabcd in the square planar SP-4 shape, as can be seen at the top of Figure 2.

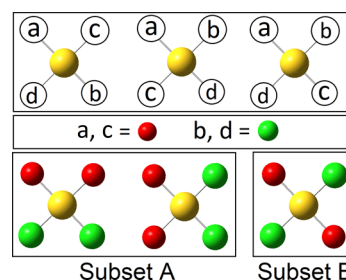


Figure 2. (top) All three and equally probable structures with four different ligands that cannot be transformed into one another by rotations. (middle) We now make a and c ligands identical and also b and d ligands identical. (bottom) The three structures are then transformed into the bottom ones: two of them cis (C_{2v}) and one of them trans (D_{2h}). Note that the two C_{2v} structures can be transformed into one another by rotation. That is why we define subset A, of the C_{2v} structure, with twice the probability of being formed when compared to subset B, of the D_{2h} structure.

Indeed, when all monodentate ligands are different, all distinct stereoisomers are equally probable under the regime of random coordination. In this article, we thus first generated all such structures for the all-different ligands situation. Then, to migrate from Mabcd to the wanted configuration Ma_2b_2 , we then paint a = c and b = d, as indicated in the middle of Figure

Table 1. Number of Stereoisomers of Two Cases of Metal Complexes^a

shape	formula	no. stereoisomers			RCR	subsets											
		total	c	a		subset	group	χ	σ	RCW	no.	subset	group	χ	σ	RCW	no.
SP-4	Ma2b2	2	0	2	2:1	A	C_{2v}	a	2	2	1	B	D_{2h}	a	4	1	1
OC-6	Ma4b2	2	0	2	4:1	A	C_{2v}	a	2	24	1	B	D_{4h}	a	8	6	1

^aShape is the polyhedral symbol according to IUPAC, with the coordination number after the dash; formula is the generic molecular formula; total is the total number of distinct stereoisomers, of which there are *c* chiral and *a* achiral ones; RCR is the random coordination ratio. All stereoisomers are classified into subsets (A or B) according to their point groups, ordered in terms of the products of their number of distinct stereoisomers, no., by their corresponding random coordination weights, RCWs. χ identifies whether all stereoisomers of the subset are chiral, c, or achiral, a; and σ is the rotational symmetry number.

2. By doing so, we obtain three new structures (Figure 2 bottom). There, we can see that the first two ended up being equivalent and different from the third one. So, we consider the first two structures as belonging to the most probable subset of structures, which we call subset A, and the remaining structure as belonging to a lower probability subset, we call subset B. As such, the stereoisomer of subset A is twice as likely to occur when compared to the stereoisomer of subset B. These numbers of structures, 2 and 1, are what we call random coordination weights, RCWs, and their proportion, 2:1, the random coordination ratio for Ma2b2. This methodology was applied to all cases to obtain the results presented in this article.

A table can now be constructed (Table 1) with these results indicating that such a composition will lead to two different complexes that can be classified into two subsets of complexes, which we call subsets A and B, shown in Table 1. The RCR for this group of complexes, also shown in Table 1, is 2:1, indicating that it is twice as probable to obtain a cis complex (C_{2v} , belonging to the A subset) than a trans complex (D_{2h} , belonging to the B subset) of general formula Ma2b2, not considering energetic effects. All complexes in this case are not chiral.

Likewise, now consider octahedral complexes of general formula Ma4b2, where M is a metal ion, and a and b are monodentate ligands. Octahedra have 12 straight edges. On the one hand, since a cis configuration means that the two b ligands must occupy one edge, there are 12 possible and equivalent arrangements around the metal ion, all leading to the same cis compound. On the other hand, octahedra have three different axes. Since a trans configuration means that the b ligands must occupy the same axis and be opposite to each other, there are three possible and equivalent arrangements around the metal ion, all leading to the same trans compound, as can be seen in Figure 3, below.

Therefore, not considering energetic effects, a cis Ma4b2 is 4 times more likely to be formed than a trans Ma4b2 if coordination happened randomly and without taking into account energetic effects. Therefore, its random coordination ratio is 4:1.

Complex Ma4b2 will be represented in this article as belonging to the set of complexes of coordination number 6, octahedral (OC-6), of general formula Ma4b2. Table 1 also shows these results indicating that such a compound will have two different and achiral complexes that can be classified into two subsets of complexes, which we call A and B and are shown in Table 1. The RCR for this group of complexes, also shown in Table 1, is 4:1, indicating that, not taking into account energetic effects, it is 4 times more probable to obtain a cis complex (C_{2v} , belonging to the subset A), than a trans complex (D_{4h} , belonging to the subset B).

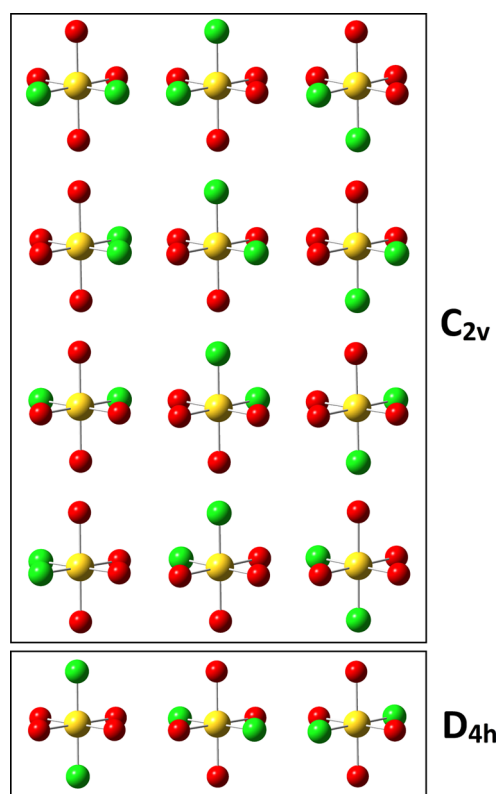


Figure 3. All 15 possible configurations that are obtained by adding two identical ligands (green) and other four identical ligands (red) to a central metal (yellow) in the octahedron shape, OC-6. Twelve of them will be cis, of point group C_{2v} , and three of them will be trans, of point group D_{4h} . Note that the D_{4h} structures can be transformed into one another by rotations. The same is true for the C_{2v} .

II. Rotational Symmetry Numbers. Rotational symmetry numbers,¹⁵ σ , the number of equivalent rotational orientations of a molecule, among several uses, are employed in kinetic models,¹⁶ in transition-state theory,¹⁷ and on the low-temperature autoignition chemistry.¹⁸

We observed that random coordination weights are related to rotations around the central metal ion, in such a manner that the group with the smallest random coordination weight will always be the one with the highest number of rotations (symmetry number). On the basis of a previous work,¹⁷ we translate this observation into the following new formula, eq 1, to calculate the rotational symmetry number σ , which we conjectured and found to be applicable to molecules and complexes with both monodentate and bidentate ligands or substituents:

$$\sigma = \frac{\prod_i (m_i!)}{\text{RCW}} \quad (1)$$

where i runs over all collections of the following species: identical atoms, identical substituents, identical ligands, equivalent coordinating atoms within a ligand or substituent, etc., in a nested manner; m_i is the number of identical species within collection i , and RCW is the random coordination weight of the complex, as defined in the present work. To verify that our conjectured formula is correct, we applied it successfully to all 2861 different structures that we considered in this work. Section IX of the [Supporting Information](#) presents a more detailed explanation together with a worked-out example.

III. Coordination Chirality. By starting with all different monodentate ligands, we can generate all different Mabdef structures that cannot be transformed into each other by rotations. The number of such structures is $6!/24 = 30$, where 24 is the number of proper rotations plus identity of the O_h symmetry point group. But now, let us address the question of chirality. These 30 structures happen to be 15 pairs of enantiomers, a fact that can be easily demonstrated using Pólya's theorem.^{19,20} As an example, [Figure 4](#) top shows two of

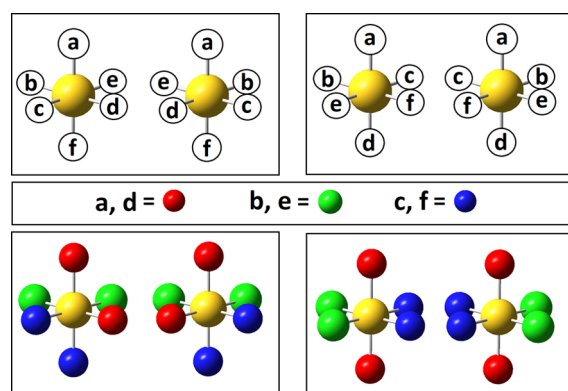


Figure 4. (top) Two pairs of enantiomers of the 15 pairs that can be formed by an octahedral complex with all different ligands. (bottom) By making ligands $a = d$, $b = e$, and $c = f$, four structures are obtained. The leftmost pair is still a pair of enantiomers, whereas the rightmost one no longer is and represents the same achiral compound, since they can be transformed into each other by a rotation.

the 15 pairs of enantiomers found for the all-different ligand complex Mabdef. By painting the ligands according to the scheme shown in [Figure 4](#) middle, we arrive at the structures of [Figure 4](#) bottom. There, it is easy to see that the two leftmost complexes still form a pair of enantiomers, while the two rightmost no longer do. Chirality can be easily discovered computationally by reflecting the geometry on a plane and comparing the resulting geometry with the ones present in the set. Once a pair is found, both geometries are grouped together as a chiral pair. This methodology is explained at length in Sections IV and V of the [Supporting Information](#).

IV. Common Shapes in Lanthanide Complexes. Because of the essentially electrostatic nature of the chemical bond between a lanthanide ion and its ligands, coordination polyhedra of several shapes may occur in such complexes. The shapes are represented by their polyhedral symbols according to IUPAC,²¹ for example, SP-4 for square or OC-6 for octahedron. The Cambridge Structural Database, CSD,²² is a repository of crystal structures, each identified by a unique

reference code comprised of six letters. We sought to identify the most common shapes that naturally occur in lanthanide complexes by selecting a small, but diverse enough, subset of complexes from the CSD with structures of high crystallographic quality (R -factor $< 5\%$) for all 15 lanthanide trications, from La to Lu, with only monodentate and bidentate ligands, and of coordination numbers ranging from 4 to 9. Accordingly, a total of 218 complexes was selected and are presented in [Table S1](#). These complexes had their coordination polyhedral shapes identified by SHAPE, a “Program for the Stereochemical Analysis of Molecular Fragments by Means of Continuous Shape Measures and Associated Tools”,^{23–25} by selecting the shape corresponding to the smallest calculated continuous shape measure for the structure. [Table 2](#) presents

Table 2. Occurrences of Coordination Numbers and Shapes for a Set of 218 Crystallographic High-Quality (R -factor $< 5\%$) Structures^a

coord no.	polyhedral symbol	shape	symmetry	set of Ln complexes	
				no.	total no.
4	T-4	tetrahedron	T_d	4	4
5	TBPY-5	trigonal bipyramid	D_{3h}	1	7
	SPY-5	square pyramid	C_{4v}	6	
6	OC-6	octahedron	O_h	16	22
	TPR-6	trigonal prism	D_{3h}	6	
	PBPY-7	pentagonal bipyramid	D_{5h}	17	
	COC-7	capped octahedron	C_{3v}	5	
7	CTPR-7	capped trigonal prism	C_{2v}	3	98
	HBPY-8	hexagonal bipyramid	D_{6h}	2	
	CU-8	cube	O_h	1	
	SAPR-8	square antiprism	D_{4d}	51	
	TDD-8	triangular dodecahedron	D_{2d}	29	
	JSD-8	snub disphenoid			
	JBTP-8	Johnson - biaugmented trigonal prism	C_{2v}	15	
	BTPR-8	biaugmented trigonal prism			
	BTTPR-8	biaugmented trigonal prism			
9	CSAPR-9	capped square antiprism	C_{4v}	6	62
	JTCTPR-9	tricapped trigonal prism (J51)	D_{3h}	38	
	TCTPR-9	tricapped trigonal prism			
	MFF-9	muffin	C_s	18	

^aComplexes selected from the Cambridge Structural Database for all 15 lanthanide trications, from La to Lu, with only monodentate and/or bidentate ligands, with coordination numbers ranging from 4 to 9. no. is the number of crystallographic structures found for each shape and symmetry. Total no. is the number of crystallographic structures found per coordination number.

the number of complexes found per shape, per coordination number for this chosen set. Therefore, [Table 2](#) only shows the shapes that were found in at least one of the 218 complexes with known high-quality crystallographic structures. [Table 2](#) thus presents the distribution of the most probable coordination numbers in lanthanide complexes, as well as the distribution of the most probable shapes per coordination number.

In this work, we enumerated the stereoisomers of all geometries proposed by SHAPE,^{23–25} in which all distances of the vertices to the center are identical.

Table 3. Number of Stereoisomers for All Cases Found in the Set of 62 Crystallographic Structures of Nine-Coordinated Lanthanide Complexes Present in Table S1, Stratified by Shape and Molecular Formula^a

		all				sets											
shape	formula	total	c	a	RCR	set	group	χ	σ	RCW	no.	set	group	χ	σ	RCW	no.
CSAPR-9	Ma(AA)2(BB)2	156	152	4	48.7:1.3:1	A	C ₁	c	1	64	146	B	C _s	a	1	64	4
						C	C ₂	c	2	32	6						
	Ma2b(AA)3	174	164	10	16.4:1	A	C ₁	c	1	96	164	B	C _s	a	1	96	10
	Ma3(AB)3	464	452	12	37.7:1	A	C ₁	c	1	36	452	B	C _s	a	1	36	12
	Ma6b3	22	12	10	12:8:1	A	C ₁	c	1	4320	12	B	C _s	a	1	4320	8
						C	C _{2v}	a	2	2160	2						
	Ma7b2	10	4	6	4:4:1	A'	C ₁	c	1	10080	4	A''	C _s	a	1	10080	4
						B	C _{2v}	a	2	5040	2						
MFF-9	Ma(AA)3(BB)	408	400	8	50:1	A	C ₁	c	1	96	400	B	C _s	a	1	96	8
	Ma2b(AA)2(BB)	2088	2070	18	115:1	A	C ₁	c	1	32	2070	B	C _s	a	1	32	18
	Ma2b(AA)3	696	682	14	48.7:1	A	C ₁	c	1	96	682	B	C _s	a	1	96	14
	Ma3(AA)3	232	222	10	22.2:1	A	C ₁	c	1	288	222	B	C _s	a	1	288	10
	Ma4b(AA)2	600	584	16	36.5:1	A	C ₁	c	1	192	584	B	C _s	a	1	192	16
	Ma5(AA)2	120	112	8	14:1	A	C ₁	c	1	960	112	B	C _s	a	1	960	8
TCTPR-9	Ma2b(AB)3	712	712	0	1	A	C ₁	c	1	12	712						
	Ma3(AA)3	31	30	1	42:1.5:1	A	C ₁	c	1	288	28	B	C _s	a	1	288	1
						C	C ₃	c	3	96	2						
	Ma3(AB)3	240	240	0	177:1	A	C ₁	c	1	36	236	B	C ₃	c	3	12	4
	Ma7(AA)	3	2	1	2:1	A	C ₁	c	1	10080	2	B	C _s	a	1	10080	1
	Ma7b2	8	4	4	2:2:1:1	A'	C ₁	c	1	10080	2	A''	C _s	a	1	10080	2
						B'	C ₂	c	2	5040	2	B''	C _{2v}	a	2	5040	2
	Ma8b	2	0	2	2:1	A	C _s	a	1	40320	1	B	C _{2v}	a	2	20160	1

^aShape is the polyhedral symbol according to IUPAC, with the coordination number after the dash; Formula is the generic molecular formula. Total is the total number of distinct stereoisomers, of which there are c chiral and a achiral ones; RCR is the random coordination ratio. All stereoisomers are classified into sets according to their point groups and ordered in terms of the product of the number of distinct stereoisomers, no., by its corresponding random coordination weight, RCW. χ identifies whether all stereoisomers of the set are chiral, c, or achiral, a; and σ is the rotational symmetry number.

The shapes enumerated in this work were: SP-4, SS-4, T-4, vTBPY-4, PP-5, SPY-5, TBPY-5, HP-6, OC-6, PPy-6, TPR-6, COC-7, CTPR-7, HP-7, HPY-7, PBPY-7, BTPR-8, CU-8, ETBPY-8, HBPY-8, HPY-8, OP-8, SAPR-8, TDD-8, CCU-9, CSAPR-9, EP-9, HBPY-9, HH-9, JTC-9, JTDIC-9, MFF-9, OPY-9, and TCTPR-9.

We also sought to investigate the geometrical characteristics of the bidentate ligands to guide our research on stereoisomers of complexes containing these types of ligands. For all selected complexes with bidentate ligands, we could not find a single case in which the two directly coordinated atoms were not adjacent in the respective coordination polyhedron. Further, consider the triangle formed by the central metal ion and the two directly coordinating atoms of the bidentate ligand. We are specifically interested in the angle opposite to the edge of the coordination polyhedron, whose vertex is located at the metal ion. This angle is called the chelate bite angle, and Table S1 of the Supporting Information contains the largest of these angles for each of the 143 selected complexes containing bidentate ligands. The median value found for this angle is 66°, the smallest of all being 34° and the largest 83°. Therefore, realistic bidentate ligands do not occupy positions in coordination polyhedra with such angles much larger than 83°.

So, in this article, we restrict the possible bidentate ligands to only those occupying adjacent positions in the coordination polyhedron and whose chelate bite angles are no larger than 90°, with the sole exception when the complex assumes a tetrahedral shape, T-4, when the four chelate bite angles are all equal to 109.47°.

V. Double Check of the Numbers of Stereoisomers.

George Pólya demonstrated in 1937 a theorem¹⁹ that fuses combinatorial analysis with group theory and that can be used to determine the number of isomers for a plurality of monodentate substitutions in a molecular scaffold. For metal complexes, this number will therefore depend on the coordination number and on the point group of the complex. Pólya's method¹⁹ thus confirms that complexes with same coordination numbers but of different shapes will possess different numbers of stereoisomers. Thus, using Pólya's theorem,¹⁹ one can easily deduce the number of stereoisomers of metal complexes coordinated with any combination of monodentate ligands, ranging from the one with all different ligands to any arrangement of repeating ligands, provided the shape of the complex is previously defined. This has already been done by other researchers for the cases of four- to eight-coordinated complexes for some shapes.²⁰

Subsequently, Haigh addressed the presence of bidentate ligands that occupy adjacent positions on the complexes, either symmetrical or asymmetrical, whose effect he showed was to reduce both the symmetry of the complex and the number of available coordination positions.^{26–28} By applying this reasoning to Pólya's method,¹⁹ Haigh was able to determine the number of stereoisomers in complexes with bidentate ligands for two different shapes of seven-coordinated complexes,^{26,28} as well as for three different shapes of eight-coordinated complexes.²⁷

In this work, we used Pólya's method¹⁹ as well as Haigh's adaptation to bidentate ligands^{26–28} as a double check to be sure that the number of stereoisomers we generated using our

methodology was correct. So, we proved to our satisfaction that our numbers are all correct despite eventual small discrepancies we found with the numbers present in the tables of one article by Haigh.²⁷ These discrepancies are all pointed out in the [Supporting Information](#), where, in one case, as an example, we present a proof that our number is correct. Finally, our results differ from those of Haigh^{26,28} for the 7-coordinated complexes with capped octahedral geometries for the cases with bidentate ligands, because he considered the possibility of such ligands with chelate bite angles up to 115.7°, which is unrealistically larger than what we found in our sample of selected complexes ([Table S1](#)).

As a final check, we further derived all numbers of stereoisomers for all shapes that had not yet been done by Krivoshei and Vvdenskii²⁰ relevant to the present work: SS-4, HP-6, PPY-6, HP-7, HPY-7, ETBPY-8, HBPY-8, HPY-8, OP-8, CCU-9, CSAPR-9, EP-9, HBPY-9, HH-9, JTC-9, JTDIC-9, MFF-9, OPY-9, and TCTPR-9; the corresponding cycle indices we derived can be found in the [Supporting Information](#) of this article.

Finally, at present we are not aware of any algebraic methods to systematically derive random coordination ratios.

VI. Tables of Stereoisomers. We determined the number of possible stereoisomers and, within it, the numbers of chiral and achiral stereoisomers, as well as the random coordination ratios. The probability of obtaining any stereoisomer belonging to a given subset if coordination happened randomly (not taking into account energetic effects) is equal to the number of stereoisomers of the subset multiplied by its RCW. We then ordered the subsets present in the random coordination patterns in terms of the probability of obtaining any stereoisomer belonging to that particular subset; subset A being the most probable one, subset B the next most probable, etc. When two subsets are equally probable but must be separated because their stereoisomers belong to different point groups, they are designated by the same letter, the first one with a prime, the next with a double prime, etc., as in B', B'', etc. [Table 3](#) contains all these data for each possible molecular formula containing monodentate and bidentate ligands for the shapes present in [Table 2](#) for the coordination number 9 only, due to size restrictions. Corresponding data for coordination numbers from 4 to 8 can be found in [Table S2](#) of the [Supporting Information](#). Finally, complete data for all possible molecular formulas with monodentate and bidentate ligands is present in [Tables S3–S36](#) of the [Supporting Information](#).

VII. Unique Notation to Represent the Configuration of a Coordination Polyhedron of a Lanthanide Complex. To completely specify the exact configuration of a lanthanide complex coordination polyhedron, together with the connections of each pair of donor atoms in the bidentate ligands, one out of a multitude of stereoisomers, we now introduce a unique notation, which is best understood by an example. Consider the crystal structure of CSD refcode NOJTEL¹⁴ of complex $[\text{Ce}(\text{fdh})_4]^-$. Our notation for this structure is $\{[\text{M}(\text{AB})_4] \text{ SAPR-8 D2 c 4 C } [1\ 4\ 8\ 3\ 2\ 6\ 7\ 5]\}$. The whole notation stays within curly brackets. First appears the generic chemical formula, in this case $\text{M}(\text{AB})_4$, indicating a metal with four asymmetrical bidentate ligands as previously explained. Then appears the shape code, as revealed by program SHAPE, in this case SAPR, square antiprism (see [Table 2](#)), followed by a dash and the coordination number, in this case 8. Then, appears the identification of chirality by either a letter c, indicating a chiral complex, or letter a,

indicating an achiral complex, followed by its symmetry number. Finally appears a capital letter representing the subset to which the stereoisomer belongs, and, within square brackets, the permutation of the primitive line vector, as described in Sections IV and V of the [Supporting Information](#). This code uniquely identifies a coordination polyhedron, together with the connections of each pair of donor atoms in the bidentate ligands, out of a myriad of possibilities.

VIII. Identification of a Given Coordination Polyhedron. On the basis of the geometries of all stereoisomers for the coordination numbers from 4 to 9 and on the shapes found and presented in [Table 2](#), we can now precisely classify any given coordination polyhedron, together with the connections of each pair of donor atoms in the bidentate ligands, of an arbitrary complex with these characteristics. First, we must identify the generic formula of the complex, for example, Ma2b2 or $\text{Ma2bc}(\text{AB})$, etc., always remembering that small letters indicate monodentate ligands and that capital letters in parentheses indicate bidentate ligands, which may be symmetric if both capital letters are the same as in (AA) or asymmetrical if they are different as in (AB). For example, for complex $[\text{Ce}(\text{fdh})_4]^-$, its generic formula is $\text{M}(\text{AB})_4$ with coordination number 8. Then, all atoms other than the ones that belong to the coordination polyhedron are deleted. The resulting structure is submitted to program SHAPE, which indicates its shape as being SAPR-8. Then, the atoms in the coordination polyhedron are labeled as in the generic code. The bidentate ligands are identified by placing a dummy atom in the midpoint between the two coordinating atoms that belong to them. All distances between the metal and the ligands are rescaled to 1. The same is done in the structures of the stereoisomers that were previously generated and are available in the [Supporting Information](#). At this point the root-mean-square deviation (RMSD) method of Marques²⁹ is determined between the structure being identified and all stereoisomers for that molecular formula and shape. The smallest RMSD found will identify which of the stereoisomers most closely matches the structure being identified. A stereoisomer identification (id) code, as introduced in the previous Section, is then associated with the coordination polyhedron. Such a code for $[\text{Ce}(\text{fdh})_4]^-$ of CSD refcode NOJTAH is $\{[\text{M}(\text{AB})_4] \text{ SAPR-8 C2 c 2 B } [1\ 3\ 4\ 8\ 6\ 7\ 2\ 5]\}$, indicating it is of general formula $\text{M}(\text{AB})_4$, of coordination number 8, of shape SAPR-8 (square antiprism), chiral, of symmetry number 2, belonging to the A subset of its random coordination pattern and of permutation $[1\ 3\ 4\ 8\ 6\ 7\ 2\ 5]$ with respect to its reference letter line vector (as described in Sections IV and V of the [Supporting Information](#)). These stereoisomer id codes will thus easily tell a researcher whether the coordination polyhedron of the compound, together with the connections of each pair of donor atoms in the bidentate ligands, is chiral or not, and to which subset of compounds (subsets A, B, C, etc.) it belongs. This procedure has been applied to all 218 selected high-quality crystallographic structures of lanthanide complexes—their respective stereoisomer id codes are presented in [Table S1](#). Section IV of the [Supporting Information](#) presents a worked-out example of the stereoisomer id code.

IX. Recognition of Chirality in Crystal Structures. Determination of the absolute configuration of a metal complex by X-ray crystallography is dependent on the ability “to identify small diffraction intensity differences between two crystal structure models of opposing chirality”.³⁰ This is far

from simple, and it usually requires a more sophisticated crystallographic procedure and, frequently, additional experiments.³⁰ Consequently, it is reasonable to assume that the absolute configurations of some of the structures we characterized in this work may not represent the true absolute configuration of the actual compound. In some cases, that could be true simply because coordination chirality was perhaps not even recognized. This is even more likely when chirality is due to the second shell of chirality,³¹ that is, due to the connections of each pair of donor atoms in the bidentate ligands.

Nevertheless, to prove the power of our methodology, we do not require that the absolute configuration reported in the CIF file be the correct one. To prove our point, we are simply accepting the absolute configuration of the structure, as it was deposited with the CCD.

Let us first consider the case of complexes $\text{Na}[\text{Ce}(\text{fdh})_4]$ (of CSD refcode NOJTEL¹⁴) and $\text{NH}_4[\text{Ce}(\text{fdh})_4]$ (of CSD refcode NOJTAH¹⁴). According to the authors, this last one, NOJTAH, contains “two crystallographically different cerium complexes (in the same unit cell) in which the coordination spheres around the cerium atoms are similar but not identical”.¹⁴ We then proceeded to classify all three structures, as they were reported, having discovered that they correspond to three different stereoisomers of $[\text{Ce}(\text{fdh})_4]^-$, as presented in Table S1. Moreover, what our classification also found out is that both structures of $\text{NH}_4[\text{Ce}(\text{fdh})_4]$ reported in the same unit cell of the CIF file are not only “similar but not identical” but also that they are actually mirror images of one another—a pair of coordination enantiomers. Figure 5, shows the

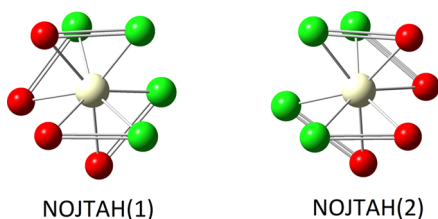


Figure 5. Reported crystallographic coordination polyhedra of the two complexes that cocrystallized in the same unit cell, of CSD refcode NOJTAH,¹⁴ showing that they form a pair of enantiomers.

coordination polyhedra of both NOJTAH structures, cocrystallized in the same unit cell, at their exact reported crystallographic geometries, together with the connections of each pair of donor atoms in the bidentate ligands, where it can be clearly seen that they constitute a pair of enantiomers.

The reported structure of the third compound, $\text{Na}[\text{Ce}(\text{fdh})_4]$, displays another chiral, but different, coordination. So, this exemplifies how easy it is to obtain different stereoisomers when there are so many possibilities and how otherwise unexpected characteristics might emerge.

Our methodology is therefore capable of flagging a coordination compound as being chiral, thus warning the crystallographer that additional experiments may be needed to completely characterize its absolute configuration.

Although chirality in metal complexes was proposed by Werner in 1899³² and proved in 1911, when he reported the successful separation of the enantiomers of $[\text{Co}(\text{en})_2\text{NH}_3\text{Cl}]\text{Cl}_2$ using d-bromocamphorsulfonate,³² it is not too frequently discussed and rarely recognized in lanthanide complexes. That may be because detecting coordination chirality is something

difficult to do without a clear methodology, as the one being advanced in this article. One of the few exceptions is the study of $[\text{Eu}(\text{TCMC})(\text{H}_2\text{O})](\text{CF}_3\text{SO}_3)_3 \cdot 2\text{CH}_3\text{OH}$,³³ where TCMC stands for 1,4,7,10-tetrakis(carbamoylmethyl)-1,4,7,10-tetraazacyclododecane, which recognized two enantiomeric pairs (four stereoisomers) appearing in the crystal. Recognition of chirality in this complex was facilitated by the fact that TCMC is octadentate, and the other ligand is a single bound water molecule.

Once the algorithm described in Section VIII is coded, it becomes relatively simple to detect chirality in lanthanide complexes. Of the 218 selected high-quality crystallographic structures of lanthanide compounds present in Table S1, 128 of them possess chiral coordination polyhedra, with almost none of the original articles mentioning this, even after sketching their respective coordination polyhedra.³⁴

Recognition of coordination chirality naturally emerges from the complete specification of the coordination stereochemistry of a lanthanide complex. Such recognition might be important, because, if the complex is both chiral and luminescent, it may emit right and left circularly polarized light with different intensities. As a matter of fact, lanthanide complexes often exhibit high values of such luminescence dissymmetry factors.^{10,35}

X. Meso Dilanthanide Compound. Dilanthanide compounds present two coordination polyhedra, one for each of the lanthanides. If both coordination polyhedra display coordinations by the same ligands, so the compound appears symmetric from the point of view of its chemical formula, it is possible that each of the coordination polyhedra is not only chiral but has opposite absolute configurations, in which case such a dilanthanide compound would actually be a meso compound, that is, a complex with two chiral stereogenic metal centers of opposite absolute configurations. An example of a dilanthanide meso compound is complex bis(μ -1,1,1,3,3,3-hexafluoropropan-2-olato)-tetra-aqua-tetrakis(1,1,1,3,3,3-hexafluoropropan-2-olato)-dieuropium of CSD refcode ERUJIM;³⁶ see Figure 6. This compound is not chiral, because it has an inversion center. We identified both of its europium coordination polyhedra as chiral and determined their stereoisomer id codes (Figure 6). We found out that they form a pair of coordination polyhedra enantiomers. As far as

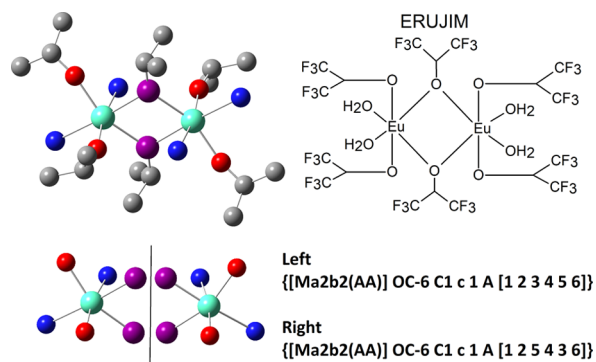


Figure 6. (top left) Skeleton of the compound of CSD refcode ERUJIM³⁶ (from the crystallographic coordinates, without hydrogen and fluorine atoms); (top right) its structural formula. (bottom left) Structures of both of its lanthanide stereogenic centers, the right one rotated so that they can be clearly seen as a pair of enantiomers. (bottom right) Identification codes of both stereogenic coordination polyhedra.

we know, this is the first time a meso dilanthanide compound has been identified as such with the lanthanide ions as stereogenic centers.

XI. High Circularly Polarized Luminescence of Chiral Complexes in Solution. Research on chiral Ln(III) complexes, emitters of circularly polarized photoluminescence, is being increasingly recognized as an emerging field, especially with regard to their applications, for example, to circularly polarized organic light-emitting diodes (OLEDs) and three-dimensional (3-D) displays.³⁷ Interest lies on the fact that Ln(III) chiral complexes exhibit exceedingly large values of the dissymmetry factor g_{lum} , defined as $g_{\text{lum}} = 2(I_L - I_R)/(I_L + I_R)$, where I_L and I_R are the left and right circularly polarized components of the emission. Among all Ln(III) complexes, CsEu[(+)-hfbc]₄ with hfbc = 3-heptafluoro-butyryl camphorate displays a g_{lum} (595 nm) = +1.38 corresponding to the europium magnetic dipole transition $^5D_0 \rightarrow ^7F_1$, which lies among the largest values observed for a single molecule.³⁵ Di Pietro and Di Bari then decided to study the geometry of CsEu[(+)-hfbc]₄ in solution, because that is where the measurement of g_{lum} is performed.³⁸ Through NMR spectra they determined that the complex static symmetry is C_4 and that the coordination polyhedron, which contains the 8-coordinated oxygen atoms in an almost perfect square antiprism (SAPR), should therefore be essentially achiral. The authors stated that a static coupling mechanism based only on directly coordinated oxygen atoms is thus insufficient to explain the exceedingly large g_{lum} value of this complex. The authors also stated that “a further source of symmetry breaking must be sought elsewhere” and proceeded to study the dynamic coupling mechanism to justify the observed g_{lum} .

Let us apply to this problem the methodology advanced in the present article. CsEu[(+)-hfbc]₄ is an 8-coordinated complex with a generic molecular formula $M(AB)_4$, since hfbc is an asymmetric bidentate ligand. From Table S25 of the Supporting Information, for shape SAPR-8 there is a C_4 subset, which is chiral. Chiral coordination in CsEu[(+)-hfbc]₄ thus comes from the C_4 arrangement of the asymmetric bidentate ligands, in the second shell of chirality³¹ due to the asymmetry of the hfbc ligand. Such chiral coordination could perhaps play a role in the static coupling mechanism of g_{lum} in CsEu[(+)-hfbc]₄, through which luminescence can become circularly polarized.³⁹

Moreover, Table S25 further indicates that, for an 8-coordinated SAPR geometry, it is possible to construct 21 chiral pairs of stereoisomers: nine of C_1 symmetry; eight of C_2 ; three of D_2 ; and one of C_4 . So, interconversion among these stereoisomers of different symmetry could perhaps occur in solution. The fact is that the authors observed the presence of three “very similar structures non- (or extremely slowly) exchanging”, with regard to which they further stated that “at the moment we are not able to provide a convincing explanation”. The possibility of factual existence of these different stereoisomers is something we believe should be considered and could perhaps play a role in the observed phenomenon.

XII. Experimental Evidence of Random Coordination Ratios. The random coordination ratios we introduced in this article indicate the probability that the actual coordination will belong to one of the subsets (A, B, C, or D) if coordination is random, not taking into account energetic effects.

Consider, for example, the case of the octahedral complex of Co(III) with three benzoylacetate asymmetric bidentate

ligands⁴⁰ of general formula $M(AB)_3$ and shape OC-6. Its random coordination ratio from Table S11, of the Supporting Information, is 3:1 with the weight 3 corresponding to the trans complex of point group C_1 (mer) and the weight 1 to the cis complex of point group C_3 (fac). The cis/trans isomerization of this complex displays an experimentally determined enthalpy variation of essentially zero. Thus, in equilibrium there should be 3 times more trans complexes than cis, a fact that has been observed in the measured cis trans equilibrium constant, found out to be very close to 1/3.⁴⁰ In like manner, in two octahedral complexes of V(III) with three asymmetrical β -diketonate ligands, V(mhh)₃ and V(mmh)₃, a mixture of 3:1 appear in the purified products.⁴¹ Other articles also present similar findings.^{42,43}

Other experimental evidence of RCRs would be that crystallized complexes in structures belonging to subsets A should be more numerous than those belonging to subsets B, and so on. Our collection of 218 complexes present in Table S1 represents a legitimate sample of lanthanide complexes chosen on the basis of crystallographic quality criterion for all 15 lanthanides, in an unbiased manner with respect to the subsets. Indeed, of the 218 high-quality crystallographic structures of lanthanide complexes present in Table S1, 129 (59.2%) belong to the A subsets, 50 (22.9%) belong to the B subsets, 34 (15.6%) to the C subsets, 3 (1.4%) belong to the D subsets, and only 2 (0.9%) belong to the E subsets, proving another benefit of the knowledge of RCRs.

XIII. Random Coordination Ratios and Theoretical Calculations. Quantum chemistry of metal complexes, of either transition-metal or of lanthanide compounds, faces the fact that a chemical structure must be assumed for the calculation to proceed. That is because, for a given formula, many different structures can be built; the question is which one of them should be chosen for the calculation? In the absence of any experimental evidence, all possible ones should be taken into account. For example, let us consider first the case of complex $[Ce(fdh)_4]^{-14}$ mentioned in the Introduction. Without any experimental knowledge of its preferred structure, all possible ones would have to be considered. First, recognize that this complex has coordination number 8 and possesses the general formula $[M(AB)_4]$. Let us first consider the most likely shape as indicated in Table 1. Therefore, assuming it will prefer a square antiprism, SAPR-8, Table S25 indicates that it will have 42 different stereoisomeric structures, all chiral, making 21 pairs of enantiomers. Next, if it prefers to arrange itself as a triangular dodecahedron, TDD-8, Table S26 indicates that it could have 51 stereoisomers, 44, making 22 pairs of enantiomers, and 7 achiral ones. Considering all remaining possible shapes: BTPR-8, HBPY-8, CU-8, ETBPY-8, HPY-8, and OP-8, there will be a total of 349 stereoisomers, of which 304 form 152 chiral pairs, and the remaining 45 are achiral ones. If the desired property is sensitive to chirality, then all 349 starting geometries must be considered in the quantum-chemical calculation. If chirality does not affect the desired property, then only one member of each pair of enantiomers must be calculated together with the achiral structures. Nevertheless, 197 starting geometries would still have to be considered for the quantum-chemical or molecular-modeling calculations. Depending on the general formula of the complex, thousands of structures would have to be considered. In the Supporting Information we present a set of .mol2 files containing the necessary connectivity to generate all such structures for an arbitrary complex, with the complete certainty

that none has been forgotten and also with complete certainty of the absence of duplicates. The fact that the structures have been classified into subsets A, B, C, etc., according to their random coordination ratios, also allows the theoretician to calculate a sample structure of each of the shapes and subsets to attempt to reduce, by energetic arguments, the number of structures to be considered. The fact that so many structures are possible has eluded many researchers, because human cognition does not easily grasp the immense number of possibilities being offered by polyhedral combinatorics. One exception is an article⁴⁴ where the authors considered 67 different starting structures for 8-coordinate catecholate and aquo complexes of uranium(IV) to ensure that “potentially viable architectures are not overlooked in the building process”.⁴⁴ The authors add that “identification of these structures can be problematic”. Indeed, and that is why the present article provides all information necessary to both generate all structures given a general formula with a guarantee of completeness, as well as to identify the full stereochemistry of the coordination polyhedron, together with the connections of each pair of donor atoms in the bidentate ligands, of a metal complex of interest in an unambiguous and definitive way.

CONCLUSION

In this article, we introduce the concept of random coordination ratios (RCRs) of metal complexes with primary focus on lanthanide compounds that display a multitude of coordination numbers, molecular formulas, shapes, and point groups. Certainly, the RCRs can also be used for transition-metal complexes, but RCRs are even more varied and elaborated for lanthanide complexes.

For example, a lanthanide complex of generic formula $M(AB)_4$, such as the one of CSD refcode CIRKET,⁴⁵ of coordination number 8 and shape TDD-8 (triangular dodecahedron) presents an RCR of 44:18:4:2:2:1:1. So, the possible stereoisomers can be collected into seven different subsets, each with a distinct point-group symmetry, respectively: C_1 , C_2 , C_3 , S_4 , D_2 , C_{2v} , D_{2d} (Table S26). As an illustration, not considering energetic effects, it is 44 times more probable to get a C_1 stereoisomer than a C_{2v} or a D_2 one.

Truly, random coordination ratios are useful for the interpretation of experimental data, such as reaction yields and spectra of complexes, because they describe the limiting situation when energetic effects do not differ among the various possible stereoisomers, such as in the case of the octahedral complex of Co(III) with three benzoylacetate ligands,⁴⁰ as described in Section XII.

Accordingly, we present, in the Supporting Information, tables that can be easily consulted, containing all RCRs and relevant information for all shapes and molecular formulas for all coordination numbers from 4 to 9.

We further present a formula that we conjectured for the calculation of symmetry numbers from the RCWs and the molecular formulas of the complexes (eq 1).

Human cognition easily overlooks the explosive number of possibilities in combinatorics. Consider, for example, the case compound bis(nicotinato-O,O')-(nicotinato-O)-diaqua-(1,10-phenanthroline)-europium(iii) dimethyl sulfoxide solvate.⁴⁶ It is not simple to grasp that the crystal structure for this compound, of CSD refcode YUXREO,⁴⁶ generic formula $Ma2b(AA)2(BB)$, and shape MFF-9, corresponds to only one of 2088 possible stereoisomers, 2070 of them being chiral and

18 nonchiral. The tables in the Supporting Information shed light on this complexity.

Furthermore, we present, in the Supporting Information, geometries for all stereoisomers of coordination polyhedra, from coordination numbers 4 to 7, that can be useful for the systematized buildup of all possible structures that would be needed to be considered, for example, for the quantum-chemical theoretical calculation of a complex of interest for which there are no experimental indications of a possible 3-D chemical structure.

From these geometries of stereoisomers of coordination polyhedra we devised an unambiguous identification of the complete stereochemistry of a given coordination polyhedron, together with the connections of each pair of donor atoms in the bidentate ligands, of a complex. Such notation can be viewed as a generalization of the R/S notational system, being naturally capable of revealing the chirality and absolute configuration of the coordination compound, a property largely overlooked in lanthanide complexes even though they often exhibit, when luminescent, high dissymmetry factors.^{10,35} So much so, that, to the best of our knowledge, we were able to recognize with our methodology, for the first time, a known dilanthanide complex as a meso compound, with both metal ions as stereogenic centers.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.inorgchem.8b01133.

Stereoisomer identification for the small sample of 218 high-quality crystallographic structures of lanthanide complexes. Stereoisomer countings, RCRs, subsets, and chirality for all generic molecular formulas and shapes, for all complexes with coordination numbers from 4 to 9 with both monodentate and symmetric and asymmetric bidentate ligands. Algebraic counting of the number of stereoisomers for the selected coordination polyhedra by means of Pólya's theorem. Symmetry numbers from the molecular formulas and RCWs (PDF)

Coordination polyhedron geometries for all cases with coordination numbers from 4 to 7 (ZIP)

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

The authors appreciate the financial support of the following Brazilian Agencies: FACEPE (Pronex), CAPES, and CNPq.

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